

Supplementary Material

To Be Is to Make Oneself One: Finite Unity as Made, and How to Test It

Scope

This supplement documents the claims of the article that have been checked in Lean 4. It is not required in order to follow the argument, and it is deliberately short: the formalization is a consistency apparatus, not a proof engine.

What a machine check establishes here is narrow, and I state it once so that it is not overread. It establishes that a stated structure is consistent, that a stated conclusion follows from stated definitions, and, in two cases below, that a stated limitation is real. It does not establish that the encoded structures are faithful to the philosophical notions they are named after. Fidelity is defended by the argument of the article and by nothing in this file.

Two of the sections below certify limits rather than results. Section C ships a theorem together with the counterexample that bounds its scope; Section D ships, as a posited field rather than a derived proposition, the one commitment the article declares as a debt. A formalization that recorded only what succeeds would be of little evidential use.

All files compile under Lean 4.28.0 (toolchain `leanprover/lean4:v4.28.0`) with no `sorry`, no added domain axiom, and no `import`: each is standalone and may be checked with `lean <file>`. Each ends with an audit reproducing, via `#print axioms`, the footprint of every theorem cited here. Lean identifiers are given in monospace; where an identifier carries a name different from the article's term, the correspondence is stated. In particular the Lean constructor `portage` is the article's *carriage*.

A. Endorsement is independent of closure

Supports §5.7, the section the article identifies as carrying its principal claim.

File: `RosenEndorsement.lean` (242 lines, 7 theorems, 0 `sorry`, 0 imports).

Two axes are encoded separately and are given no field in common. Closure to efficient causation reads a catalyst graph alone:

```

structure Network where
  requires : List Nat
  produces : List Nat

def RosenClosed (net : Network) : Prop :=
  net.requires.all (fun c => net.produces.contains c) = true

```

Endorsement reads an energetic profile alone:

```

structure EnergyProfile where
  margin : Nat
  reconfig_cost : Nat
  energy_external : Bool

def EndogenousEndorsement (p : EnergyProfile) : Prop :=
  p.margin ≥ p.reconfig_cost ∧ p.margin > 0

```

The field `energy_external` is present and is deliberately idle. Energetic dependence on a milieu is universal, as §5.6 argues, so it must not sort, and no theorem below turns on it.

Four witnesses inhabit the four cells of closure $\times$ endorsement (`square_inhabited`), so neither axis is a function of the other. The decisive one is a network that produces every catalyst it requires and carries no store: the flow reactor of §5.7.

```

theorem closure_does_not_entail_endorsement :
  ¬ (∀ e : Entity, closed e → endorses e)

theorem endorsement_separates_within_closure :
  closed organism ∧ closed selfCatalyzingFlow ∧
  verdict organism = Verdict.self ∧
  verdict selfCatalyzingFlow = Verdict.carriage

```

The second is the pair the argument needs. `organism` and `selfCatalyzingFlow` are both closed, so Rosen’s criterion cannot tell them apart, and the endorsement criterion assigns opposite verdicts. Note that this is not the organism/whirlpool pair of §5.4, which closure already separates and which therefore establishes only the weaker claim that cost is present where closure is silent.

Functional independence is then stated directly: entities sharing an energetic profile share a verdict, whatever their catalyst graph.

```

theorem verdict_ignores_closure (e1 e2 : Entity)
  (h : e1.profile = e2.profile) : verdict e1 = verdict e2

```

On the strength of the rendering. Closure is encoded minimally, as internal production of every required catalyst. Richer (M, R) structure adds constraints, and adding

constraints cannot manufacture the witness that matters here, which is a closed entity that fails to endorse. The independence therefore holds a fortiori for any faithful or richer rendering, provided closure remains silent about energy storage, which it does.

B. The verdict is a function of the cost profile alone

Supports §5.8, at the point identified there as the account’s greatest exposure.

File: `gradient.lean` (91 theorems, 0 sorry, 0 imports); the result is at line 602.

```
structure CostProfile where
  self_cost : Nat
  host_cost : Nat

def regimeOf (p : CostProfile) : CostRegime :=
  if p.self_cost > 0 then CostRegime.closure
  else if p.host_cost > 0 then CostRegime.portage
  else CostRegime.aggregate

theorem cost_profile_determines_regime (p1 p2 : CostProfile)
  (h_self : p1.self_cost = p2.self_cost)
  (h_host : p1.host_cost = p2.host_cost) :
  regimeOf p1 = regimeOf p2
```

The article claims that the location of the irreversible trace does not vary with the observer’s carving, and concedes that the claim is empirical in form. A formalization cannot settle an empirical claim. What it settles is the prior question of whether the criterion *as encoded* contains any discriminant other than the cost split, and it does not: the regime is a function of the pair (`self_cost`, `host_cost`), so two entities agreeing on that pair agree on their regime whatever else differs between them.

The proof is a rewriting under the definition, and I do not present it as more than that. A reader is entitled to observe that a profile type with two numeric fields will not exhibit a third parameter. The point is what the type omits: substrate, scale, topology, label, and observer are not fields of `CostProfile`, so no theorem in the file can appeal to them, and this is checkable rather than asserted.

C. The first branch, and the counterexample that bounds it

Supports §5.3 and the first question of Figure 1.

File: `7autoOnto.lean` (40 theorems, 0 sorry, 0 imports); the result is at line 854.

```

theorem no_third_regime (sys : FiniteSystem) (start : Fin sys.states) :
  classifyTrajectory sys start = Regime.dissolves ∨
  classifyTrajectory sys start = Regime.closure

```

A companion result, `closure_has_cycle`, bridges the two: if a trajectory does not dissolve then it revisits a state, and every intermediate state has positive margin.

Two restrictions belong with the theorem, and both are recorded in the source rather than left implicit.

First, the scope. `FiniteSystem` presupposes a finite discrete state space (`Fin states`) and deterministic succession. The article’s “every exposed finite being” becomes, in the formalization, “every system on a finite discrete state space”. A being bounded in margin but with a continuous configuration space is not covered; the corresponding statement there would fall under Poincaré recurrence, and I have not established it. The counterexample is explicit: on $\mathbb{N}$ with successor $n \mapsto n + 1$ the trajectory never revisits a state, and neither disjunct holds.

Second, what the theorem does not give. It yields recurrence, not endorsement. That a trajectory returns to a state says nothing about which margin paid for the return, and the trichotomy of §5.3 is settled at the second branch, not this one.

A further result records the status of the third regime. Under conservation of the maintenance cost together with its strict positivity, no entity has both shares null:

```

theorem aggregate_impossible (e : FiniteEntity) : ¬ noRegeneration e

```

Caution on use. This concerns the distribution of a cost already taken as given, and it is the formal counterpart of the article’s claim in §5.3 that carriage and aggregate are privations rather than positive kinds. It is not a claim about the stone, and it does not bear on the equivocation §2.5 refuses, where the drain of an aggregate is distinguished from an expenditure that buys continuation.

D. A declared debt, mechanized

Supports §7.3.

File: 7autoOnto.lean, line 287.

```

class FiniteExposed (α : Type) where
  margin      : α → Nat
  drain       : α → Nat
  drain_pos   : ∀ a, 0 < drain a

```

The strict positivity of the drain is a field of the class, not a proposition derived within it. This is the formal shape of the debt §7.3 declares: the article presents

the drain as a posit about our irreversible world, and the development enters it as a constraint rather than proving it. Every result carrying the class inherits that status, including the exhaustion theorem `generic_exhaustion` proved for all `FiniteExposed` types.

I include this section because a supplement certifying only successes would leave the most checkable of the article’s four debts uncorroborated.

E. The separating model of immanent necessity

Supports §6.2.

File: INu_Necessity.lean (352 lines, 13 theorems, 0 `sorry`, 0 imports).

The argument of §6 turns on the alternative being coherent: if the contingent act could not be constructed without contradiction, the necessity claimed for the self-grounded act would be obtained by definition.

Two act types are opposed. `SelfGroundedAct` has an endogenous margin and no external ground; `act_is_necessary` establishes that persistence requires the cost to be payable on the proper margin, and `inaction_is_dissolution` that inaction is not a stable state. `ContingentAct` carries an `external_ground` that compensates the cost; `contingent_survives_without_economy` and `contingent_waste_not_existential` establish that for it the cost of renewal is optional and waste is not existential. `contingency_needs_ext` shows that a non-necessary act would require an external ground, and `self_ground_suffices` that self-grounding excludes one.

What this certifies is joint constructibility: the two acts coexist in one consistent development, so the necessity of §6.2 is conditional on self-grounding rather than analytic. It does not certify that “self-grounded” as encoded captures self-grounding as the article uses the term.

F. What is not formalized

Stated for completeness, so that the unlisted is not taken to be unformalizable.

The argument of §2 is not mechanized and is not a candidate for mechanization: that finitude entails exposure, and that the ideal crystal removes exposure while retaining finitude, are conceptual claims about the notions involved and not consequences of a stated structure.

The independence result of Section A above is proved for a minimal rendering of closure. A rendering with full (M, R) replacement structure has not been carried out, and the article’s claim that the independence survives it is argued rather than checked.

The material on the intermediate regime and its parameter of modal renewal,

present in earlier presentations of this framework, has no referent in the present article and is not included here.

Availability

The Lean sources are supplied with this submission. The public repository address is withheld to preserve anonymity and will be supplied on acceptance.